CrowdSense: Real-Time Crowd Formation Detection and Risk Assessment for Smart City Surveillance

By: Yigal Meshulam & Yuval Wilf

Advisor: Dr. Yehudit Aperstein

Abstract

CrowdSense is a near real-time framework for detecting emergent crowd gatherings in CCTV-like video and generating near-term safety risk assessments that support operational decision-making in smart city control settings. The system is designed for periodic sampling (e.g., 1 frame per 15 seconds per feed), suitable for control room environments where operators monitor multiple camera streams. The system uses density-map-based crowd counting to estimate human presence under occlusion and scale variation, avoiding brittle person detection and tracking in congested scenes. Each frame is converted into a crowd density map (via DM-Count) and summarized into a compact set of "gathering signals" capturing crowd size and spatial structure (e.g., total count, hotspot concentration, spread, and coarse zone occupancy).

For risk assessment, CrowdSense introduces a two-phase reasoning layer combining deterministic rules with generative AI. Phase 1 uses a configurable rule engine to perform single-frame analysis, producing an operator-style status label (Sparse, Moderate, Crowded, Heavily Crowded) with a short justification derived strictly from the extracted signals. When the status reaches Moderate or above, Phase 2 is triggered: an LLM receives a short time-ordered sequence of recent frame summaries (up to 6 "latest events") and outputs a risk assessment report that prioritizes escalation patterns (e.g., surge vs. compression/locking) and suggests operator checks and monitoring steps. **Overall, CrowdSense turns density maps into a standardized, interpretable feature interface and uses a gated two-phase (rules + LLM) reasoning layer to deliver operator-style crowd state and short-horizon escalation explanations.** By combining density-driven measurement with deterministic classification and interpretable LLM reporting over short-term temporal context, CrowdSense aims to provide early-warning cues before crowd build-up evolves into hazardous conditions.

---

1. Introduction

1.1 Motivation and Problem Statement

Large public gatherings in smart cities (e.g., concerts, demonstrations, sporting events) can rapidly shift from organized activity to high-risk situations, including overcrowding and violent escalation. City control rooms often rely on manual monitoring across many camera feeds (an approach that is slow, subjective, and difficult to scale) creating a need for operational early-warning capabilities that detect emerging gatherings and support timely, proportionate intervention when risk indicators begin to rise.

CrowdSense is motivated by two operational needs: (1) detecting when a gathering is forming so relevant authorities can assess legitimacy via policy and external channels, and (2) assessing near-term crowd risk (e.g., over-density, stalled flow, turbulent motion) to guide response. Technically, explicit person detection and tracking are fragile in surveillance conditions due to occlusions, perspective effects, and dense congestion. Density-based crowd analysis offers a practical alternative by estimating both crowd size and spatial concentration from continuous density maps, even when individuals cannot be reliably separated.

1.2 Research Question

Can emergent visual signatures in CCTV surveillance be used to detect the formation of spontaneous gatherings and assess their escalation into dangerous situations?

1.3 CrowdSense Overview and Contributions

CrowdSense is a near real-time, periodic-sampling framework designed to support operational monitoring by converting CCTV frames into interpretable signals about crowd formation and short-horizon escalation risk. Instead of relying on explicit detection and tracking of individuals, CrowdSense adopts a density-based representation of human presence: each frame is mapped to a continuous crowd density estimate, enabling robust measurement under occlusion and congestion. The density output is then summarized into compact “gathering signals” that capture both volume (how many people) and structure (how concentrated, how spread, and where the crowd is localized). This standardized representation enables consistent monitoring across camera views and provides a structured input for downstream reasoning.

Building on these signals, CrowdSense performs risk reasoning in two stages. First, a single-frame analysis produces an operator-style status label (e.g., Sparse to Heavily Crowded) with a short justification grounded in extracted indicators. When the status indicates a meaningful gathering, the system then analyzes a short sequence of recent frame summaries to identify escalation patterns such as rapid influx, increasing concentration/compaction, or constrained crowd geometry. It outputs a structured risk assessment report that highlights the primary drivers behind the assessed risk level and suggests operator checks and monitoring steps, prioritizing interpretability and operational usefulness.

Main Contributions:

CrowdSense turns density maps into a standardized, interpretable feature interface and uses a gated two-phase (rules + LLM) reasoning layer to deliver operator-style crowd state and short-horizon escalation explanations.

1. A density-driven pipeline for gathering detection that avoids fragile per-person tracking in congested scenes

2. A standardized set of “gathering signals” derived from density maps for both per-frame interpretation and temporal trend analysis
3. A two-phase decision layer combining a configurable rule engine (Phase 1) with LLM-based temporal analysis (Phase 2) to produce operator-facing status and risk reports with explicit drivers and actionable guidance
4. A configuration-first design enabling scenario-dependent threshold calibration across diverse deployment contexts

1.4 Target Applications

- Public event monitoring (concerts, festivals, sports venues)
- Transportation hub management (stations, airports, terminals)
- Retail analytics (store occupancy, queue detection)
- Urban planning studies (pedestrian flow analysis)
- Emergency response support (evacuation monitoring)

---

2. Literature Review and Existing Approaches

Detecting human gatherings in public spaces is central to urban safety, public health, and event monitoring. In real surveillance environments, explicit person detection and multi-object tracking often fail under the very conditions that matter most - partial occlusion, scale variation, and heavy congestion. As a result, a major line of work has shifted toward density-based crowd analysis, where people are modeled as a continuous spatial density distribution rather than as independently detected objects [2]. Density maps enable estimation of both crowd size and spatial concentration even when individuals cannot be reliably separated, forming the basis of modern crowd counting and downstream scene reasoning [1,2].

2.1 Density-Based Crowd Counting

Most modern crowd counting methods rely on density map regression, where convolutional networks predict a continuous density field whose integral yields the crowd count. CSRNet introduced a strong baseline for highly congested scenes by showing that dilated convolutions can produce accurate density estimates under complex perspective and occlusion [2]. DM-Count advanced this paradigm by framing learning as distribution matching between predicted density and point-level annotations, resulting in sharper localization and strong counting performance across standard benchmarks [1]. In operational monitoring settings, these methods provide a robust measurement layer that can continue to function where detection and tracking degrade.

Limitation for safety monitoring: crowd counting models primarily output counts (and sometimes density fields) but do not directly answer higher-level operational questions such as whether a gathering event is forming or whether the crowd’s state is escalating toward risk. Bridging this gap requires additional structure on top of density estimation.

2.2 From Density Maps to Gathering Detection

Density maps also support localization and hotspot reasoning: peaks and local maxima can act as proxies for head locations or clustered subregions [1,2]. This enables downstream tasks such as estimating per-region occupancy, identifying crowded subareas, and tracking where density concentrates over time - capabilities that are useful for defining "gatherings" without requiring explicit per-person tracks. In particular, coarse zone-level summaries and hotspot dominance indicators can be monitored consistently across camera feeds and mapped naturally to control-room workflows.

Key gap: while prior density-based work provides raw signals (counts and spatial density), it often stops short of offering an operational definition of gathering formation that is immediately usable in control-room settings (e.g., standardized indicators and zone summaries that support rapid interpretation across many cameras)..

2.3 Crowd Risk, Congestion, and Safety Indicators

A distinct body of literature emphasizes that the main hazard in crowd disasters is not merely the presence of a crowd, but the transition into dangerous dynamics as density rises and movement becomes restricted. Empirical and modeling studies show that extreme density in confined geometries can produce stop-and-go waves and turbulent motion that precede crush incidents [3]. Building on this principle, Bek and Monari proposed a video-based crowd congestion level that explicitly couples local density with reduced motion to quantify risk in surveillance footage [4]. Many related approaches incorporate motion cues (e.g., optical flow or tracklet statistics) to detect stagnation and flow constraint, but these signals can degrade under occlusion and often require scene-specific calibration. Overall, this literature motivates risk indicators that jointly reflect crowd volume, spatial compaction, and constraints on flow - signals that are more aligned with safety outcomes than count alone.

Limitation: many safety indicators require either explicit motion estimation or scene-specific calibration, and they are not always presented in a form that supports rapid operator interpretation across multiple live feeds

2.4 Learning-Based Anomaly and Risk Prediction from Crowd Features

More recent work explores learning-based models that operate on crowd-level features to detect or predict hazardous conditions. The SIMCD framework, for example, uses simulated crowd data with severity and abnormality labels to train both unsupervised anomaly detectors and supervised temporal models (e.g., sequence models) for early warning of dangerous configurations [5]. This line of work supports the broader claim that short histories of crowd features - density, speed, and directional consistency - can be sufficient to forecast escalation.

Practical challenges: in real surveillance deployments, labeled incident data is scarce, environments vary widely, and purely model-based risk scores can be difficult to trust without interpretable drivers. This motivates approaches that combine (i) robust feature extraction, (ii) temporal reasoning over short windows, and (iii) outputs that explain why a risk assessment was produced.

2.5 Summary of the Gap and Positioning of CrowdSense

Across these strands, prior work has established how to (1) obtain reliable density maps from single frames [1,2], (2) link high density and constrained movement to critical crowd dynamics [3,4], and (3) use crowd-level features for anomaly detection and forecasting [5]. However, an operational monitoring system still needs a cohesive bridge between these components: a standardized, camera-agnostic feature interface derived from density maps, plus short-window temporal reasoning that produces interpretable, operator-facing outputs. Therefore, the open engineering gap is not density estimation itself, but turning density into a monitoring representation and escalation logic that can be applied consistently across diverse scenes and trusted by operators.

CrowdSense is positioned to connect these pieces in a single workflow: a density-map backbone (DM-Count) provides robust per-frame spatial measurements [1], which are summarized into geometry-aware indicators inspired by safety literature [3,4] and then used for short-horizon escalation reasoning consistent with feature-based forecasting approaches [5]. The emphasis is not only on predicting risk, but on making the assessment traceable to concrete measurable factors (e.g., rising density, increasing hotspot dominance, compaction, and stagnation), enabling practical early-warning support in smart city surveillance settings

---

3. System Design

3.1 Architectural Philosophy

CrowdSense follows a layered architecture with dependency inversion, where high-level policies (domain logic) do not depend on low-level details (infrastructure implementations). This separation keeps operational logic stable while allowing underlying components - such as the density estimator, storage backends, or LLM provider - to be substituted without changing core behavior. The architecture is intentionally modular to support incremental deployment in control-room environments and to simplify testing, maintenance, and extension.

A central design goal is configuration-first adaptability. Crowd monitoring thresholds and alert policies are inherently scenario-dependent: what constitutes “Moderate” in a retail store differs from a stadium concourse or a city square. CrowdSense therefore externalizes key parameters - status thresholds, concentration alerts, gating rules, and response templates - into configuration files, enabling rapid calibration per camera, per location, or per operational policy without code changes. This approach supports broader adoption across varied deployment contexts while preserving consistent system semantics and auditability.

This design enables:

- Technology substitution without domain changes
- Independent testing of each layer
- Clear boundaries for maintenance and extension

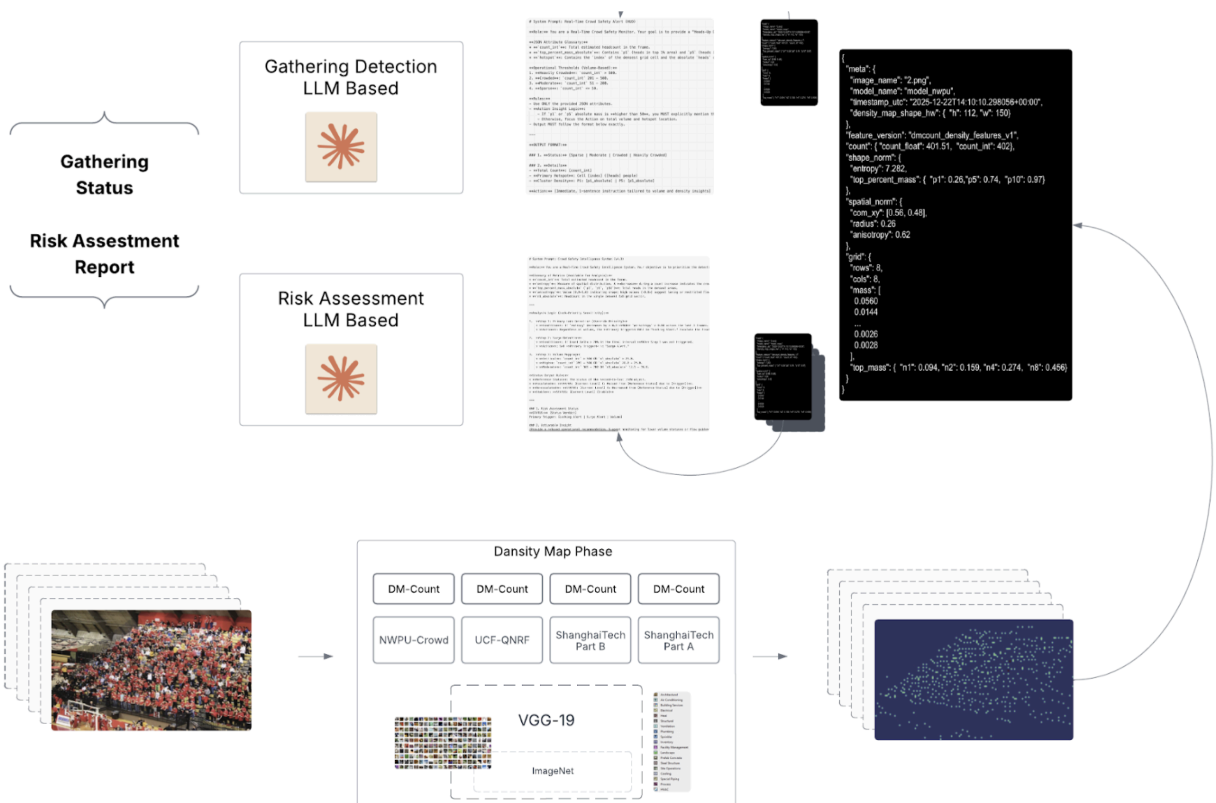
- Scenario-dependent calibration through configuration (thresholds, policies, and actions) without code modification

3.2 System Architecture Overview

The system is organized into four primary layers, as follows:

Layer	Responsibility	Key Principle
Presentation	User interaction, display, event capture	UI-agnostic coordination
Service	Pipeline orchestration, error handling	Protocol-based abstraction
Analysis	Alert generation, risk assessment	Dual-mode (deterministic + generative)
Infrastructure	Model inference, I/O operations	Concrete implementations

3.3 Architecture High Level Diagram



3.4 Processing Pipeline

CrowdSense follows a modular Sense–Detect–Assess flow: video is periodically sampled into frames, each sampled frame is converted into a crowd density representation, the density map is summarized into compact “gathering signals,” and these signals are consumed by a gated two-phase reasoning layer. Phase 1 applies a deterministic, configurable rule engine for immediate per-frame status classification. When the status indicates a meaningful gathering (≥

Moderate), Phase 2 is triggered: a temporal analyzer (LLM) consumes a short, time-ordered window of recent frame summaries and produces an escalation-oriented risk assessment report. The outputs are formatted for operator consumption (status, drivers, and suggested checks/actions) and can be integrated into a dashboard interface.

This separation enables:
robust visual measurement under occlusion and congestion
a standardized, camera-agnostic feature interface for downstream logic
fast, deterministic status reporting via configurable thresholds and policies
interpretable escalation analysis over short temporal windows, invoked only when warranted

The end-to-end pipeline processes frames through sequential transformation stages, from validation and preprocessing through density inference, feature extraction, and alert generation.

Stage	Input	Output	Purpose
Validation	File path	Metadata object	Verify format, size, existence
Preprocessing	File path	Normalized tensor	Convert to model-ready format
Inference	Tensor (C,H,W)	Density map (H/8,W/8)	Generate spatial density
Statistics	Density map	Feature dictionary	Extract interpretable metrics
Feature Extraction	Statistics	Structured JSON	Package for downstream use
Alert Generation	Features	Status + Action	Classify and recommend

---

4. Methodology

4.1 Density Estimation Backbone

CrowdSense uses a density-map crowd counting model as its perception backbone. In this paradigm, the model predicts a continuous density field over the image rather than explicitly detecting and tracking each person, which is advantageous in crowded scenes where individuals are difficult to separate.

4.1.1 DM-Count Model

The implementation is based on DM-Count (Distribution Matching for Crowd Counting) with a convolutional feature extractor (VGG-style backbone) and a density prediction head; pretrained weights can be used and optionally adapted to the deployment domain via fine-tuning on standard crowd-counting datasets.

4.1.2 Why Density Maps?

Approach	Pros	Cons
Direct Regression	Simple output	No spatial info, poor gradients
Detection-based	Precise localization	Fails in dense crowds

Density Estimation	Spatial + count, robust	Requires ground truth maps

DM-Count uses density estimation with Optimal Transport loss, achieving state-of-the-art accuracy while preserving spatial information essential for crowd management applications.

4.1.3 Output Characteristics

- Resolution: 1/8th of input (due to backbone downsampling; prediction is produced at reduced stride)
- Value range: Non-negative floats (ReLU activation)
- Interpretation: Each pixel represents local crowd density; sum equals total count
- Normalization: Optional probability distribution (sum = 1) for statistical analysis

4.2 Density-to-Feature Representation

For each frame, the predicted density map is converted into a standardized set of “gathering signals” serialized as a compact JSON record. The feature schema is designed to capture both crowd size and crowd structure, and to be usable for single-frame interpretation as well as temporal trend analysis.

4.2.1 Feature Categories

The main feature families are:

Category	Features	Purpose
Volume (how many)	Total count, integer round	High-level status reporting and alerting
Concentration (how packed)	Entropy, top-1%/5%/10% mass	Quantify crowd concentration in densest regions
Spatial geometry (where/shape)	Center of mass, spread radius, anisotropy	Indicate corridor-, queue-, or lane-like configurations
Zone/grid summaries	8×8 cell mass, top-k cell mass, hotspot	Capture localized crowding at operational resolution

4.2.2 Feature Definitions

Feature	Description	Application
Count	Sum of density values	Primary crowd size metric
Entropy	Shannon entropy of distribution	Uniformity measure
Top-P Mass	Fraction of mass in top P% pixels	Concentration indicator
Center of Mass	Weighted centroid (x,y)	Crowd location tracking
Radius	Spatial spread from centroid	Crowd extent measure
Anisotropy	Directional asymmetry [0-1]	Shape indicator (0=circular, 1=linear)
Grid Mass	Per-cell density in 8×8 grid	Zonal analysis
Hotspot	Cell with maximum density	Critical area identification

Precise Definitions:

- Entropy: Given a density map D, compute the normalized probability distribution $p = D / \Sigma(D)$. Shannon entropy is $H = -\Sigma(p \cdot \log(p))$ for all pixels where $p > \varepsilon$ (ε = 1e-12 to handle zeros). Higher entropy indicates uniform spread; lower entropy indicates concentration.
- Top-P Mass: Sort all pixels of p in descending order. Top-P% mass is the cumulative sum of the top P% of pixels (by pixel count). For example, P1 captures what fraction of the total crowd is in the densest 1% of the image area - high P1 indicates extreme localization.
- Grid Mass: Partition the density map into an 8×8 grid (64 cells). For each cell, sum the normalized density values. Top-N metrics (n1, n2, n4, n8) report the cumulative mass of the N highest-density cells, indicating how concentrated the crowd is across coarse zones.

Each per-frame record can be logged into a lightweight time series ("latest events"), enabling risk reasoning over short temporal windows without storing full video

4.2.3 Feature Persistence and Schema

To support temporal analysis and operational reproducibility, CrowdSense serializes extracted features to persistent JSON files stored in datasets/stats/. Each file follows the dmcount_density_features_v1 schema, containing metadata (image name, model checkpoint, timestamp), volume metrics (count), concentration indicators (entropy, top-P mass), spatial geometry (center of mass, radius, anisotropy), and grid-based zone summaries. This standardized format enables Phase 2 temporal reasoning by providing a 6-frame sliding window of recent feature records without requiring raw video storage - a design choice that reduces storage overhead and supports privacy-conscious deployment. The schema includes both normalized features (for cross-scene comparison) and absolute metrics (for operational thresholds), with versioning to ensure backward compatibility as the feature set evolves. Full schema documentation is maintained in the repository (datasets/README.md).

4.3 Alert and Risk Assessment System

CrowdSense implements a two-tier alert system combining deterministic rules with genAI.

4.3.1 Scenario-Dependent Configuration

Crowd density thresholds are inherently context-dependent. What constitutes a "crowded" gathering varies dramatically across scenarios:

Scenario	Sparse	Moderate	Crowded	Critical
Road Intersection	<10	10-25	25-50	>50
Retail Store	<20	20-50	50-100	>100
Concert Venue	<100	100-500	500-2000	>2000

Scenario	Sparse	Moderate	Crowded	Critical
Stadium	<500	500-2000	2000-10000	>10000

A fixed threshold system would generate false alarms in large venues while missing dangerous conditions in confined spaces. CrowdSense addresses this through a configuration-first design where all detection thresholds and response actions are externalized to configuration files.

The system uses two primary configuration files:

1. Rule Engine Configuration (gathering_rules.json)

- **Count Thresholds**: Count boundaries for Sparse/Moderate/Crowded/Heavily Crowded status
- **Density Alert Threshold**: Trigger level for high-concentration warnings
- **Action Templates**: Status-to-action message mappings for normal and high-density scenarios

2. LLM Prompt Configuration (risk_assessments_prompt.txt)

- **Volume Thresholds**: Count and density levels for MODERATE/HIGH/CRITICAL classification
- **Lock Detection Parameters**: Entropy decrease and anisotropy thresholds for crowd compression detection
- **Surge Detection Parameters**: Count delta thresholds for rapid crowd growth alerts

4.3.2 Phase 1: Single-Frame Status Detection

Phase 1 provides a per-frame operational summary intended for fast operator consumption. The deterministic rule engine receives a single-frame JSON feature record and returns a fixed-format output containing: (i) a discrete crowd status label (Sparse/Moderate/Crowded/Heavily Crowded), (ii) detailed metrics including total count, primary hotspot location (e.g., “Cell R3C5”), and cluster density indicators (P1, P5), and (iii) an initial suggested operator check.

The labeling logic is threshold-based, with thresholds treated as configurable policy parameters (e.g., per camera or per environment). The rule engine also triggers high-density alerts when concentration metrics (P1, P5) exceed the configured density_alert_threshold, ensuring localized compression is not hidden by an “average” crowd count.

4.3.3 Phase 2: Temporal Risk Assessment

Phase 2 is triggered only when Phase 1 classifies the scene as Moderate, Crowded, or Heavily Crowded (i.e., status ≥ Moderate). The LLM receives a time-ordered sequence of up to 6 recent feature records (“latest events”) and produces a structured risk assessment report with immediate operator actions.

Temporal risk reasoning prioritizes escalation mechanisms consistent with crowd safety theory. The prompt implements a hierarchical trigger logic where high-risk patterns override simple volume-based alerts:

- **Compression-like / flow-constraint indicators (highest priority):** Triggered when entropy decreases by >0.3 AND anisotropy exceeds 0.80 across the last 3 frames. These patterns *suggest* constrained flow or crowd compaction, but do not confirm physical crush conditions-operators should verify via camera review or on-ground personnel before escalating response.
- **Surge patterns:** Triggered when count delta exceeds 20% in the final interval, identifying rapid crowd growth even if density is not yet extreme.
- **Volume patterns (baseline):** Triggered when count_int >100 OR n1_absolute >12.0, with severity levels:
  - I. CRITICAL: count >500 OR n1_absolute >25.0
  - II. HIGH: count 201-500 OR n1_absolute 20.0-25.0
  - III. MODERATE: count 101-200 OR n1_absolute 12.1-19.9

Note: The thresholds listed above are example values suitable for typical outdoor surveillance scenarios. All volume, compression, and surge detection parameters are configurable via the LLM prompt configuration file (risk_assessments_prompt.txt), enabling operators to calibrate risk triggers for specific deployment contexts (e.g., confined spaces, large venues, transportation hubs).

The output report includes: (i) a relative status trend (stable/escalating/de-escalating), (ii) the primary trigger category, (iii) concise driver explanations grounded in the observed feature trends, (iv) suggested operator checks and monitoring steps, and (v) a compact summary table of recent key indicators for transparency.

4.3.4 LLM Reliability and Guardrails

To ensure predictable and safe operation, Phase 2 implements several constraints on LLM behavior:

- **Output constraints**: The LLM is instructed to output a fixed JSON schema with a constrained risk level enum (MODERATE/HIGH/CRITICAL). The prompt explicitly prohibits scene invention or speculation beyond the provided feature data.
- **Validation and fallback**: The system validates the LLM response against the expected schema before displaying to operators. If the response is malformed, times out, or fails API limits, the system falls back to Phase 1 status with a "Phase 2 unavailable" indicator.
- **Reproducibility and cost control**: Responses are cached by content hash of the input feature sequence, ensuring identical inputs produce consistent outputs and reducing redundant API calls during stable scenes.

4.4 Terminology Reference

To ensure consistent interpretation across the paper, the following table maps the two-phase terminology to their operational meaning:

Term	Phase	Meaning
Crowd State	Phase 1	Snapshot assessment of volume and spatial structure
Sparse	Phase 1	Low crowd density, normal operations
Moderate	Phase 1	Notable gathering, increased monitoring warranted
Crowded	Phase 1	High density, active management recommended
Heavily Crowded	Phase 1	Critical density, intervention protocols triggered
Risk Level	Phase 2	Short-horizon escalation likelihood based on temporal trends
MODERATE	Phase 2	Elevated risk indicators, enhanced monitoring
HIGH	Phase 2	Significant escalation pattern detected
CRITICAL	Phase 2	Severe risk indicators, immediate attention required

Phase 1 labels describe the *current state* (what the crowd looks like now), while Phase 2 labels describe the *risk trajectory* (how the situation is evolving).

---

5. Implementation

5.1 Technology Stack

Category	Technology	Purpose
Language	Python 3.10 (3.8+ compatible)	Primary implementation language
Deep Learning	PyTorch	Model inference framework
Image Processing	Pillow, OpenCV	Image I/O and visualization
Numerical	NumPy, SciPy	Statistical computations
UI Framework	Tkinter	Desktop interface
LLM Integration	LiteLLM	API abstraction for language models
Local LLM	Ollama (qwen2.5:14b)	Local inference option for offline deployment

5.2 Configuration System

The configuration-first design allows operators to tune the system for their specific deployment context without code changes, while maintaining consistent alert logic across scenarios.

Rule Engine Configuration controls the deterministic classification tier through a JSON file containing: status thresholds (Sparse/Moderate/Crowded/Heavily Crowded), density alert thresholds for concentration warnings, and action message templates for normal and high-density scenarios.

LLM Prompt Configuration controls the generative analysis tier through a text file defining: volume thresholds for MODERATE/HIGH/CRITICAL risk levels, lock detection conditions (entropy decrease + anisotropy), and surge detection parameters (count delta percentage).

5.3 Performance Considerations

5.3.1 Computational Profile

Stage	Typical Latency	Bottleneck
Validation	<5ms	File I/O
Preprocessing	~20ms	Image decoding
Inference	~30-50ms (GPU)	Neural network forward pass
Statistics	<10ms	NumPy operations
Rule Engine	<1ms	Threshold comparisons
LLM (optional)	50-75s	Network + API processing

5.3.2 Optimization Strategies

- **Hardware acceleration**: automatic selection (CUDA when available, otherwise Apple MPS on macOS, otherwise CPU)
- **Response caching**: LLM results cached by content hash of the input feature sequence
- **Lazy loading**: model loaded on first use, not at startup
- **Background threading**: long operations run off the UI thread
- **Gated invocation**: temporal risk reporting invoked only when non-trivial crowd is detected

5.3.3 Deployment Considerations

The methodology is designed for near real-time deployment at periodic sampling rates (e.g., 1 file per 15 seconds per feed), suitable for control room settings where operators monitor multiple camera streams. Practical constraints include: runtime depends on frame resolution and model throughput; crowd/risk thresholds may require calibration to local camera geometry and operational policy; and Phase 2 latency/cost can be significant. CrowdSense addresses this with temporal decoupling: Phase 1 provides sub-second dashboard status, while Phase 2 runs as a gated, asynchronous early-warning assessment triggered only when status ≥ Moderate

---

6. Experiments and Evaluation

We evaluate CrowdSense across three dimensions: (1) backbone capability, reported using published DM-Count benchmark results, (2) computational performance of the end-to-end pipeline, and (3) behavioral validation of the alert and risk assessment system (Phase 1 rules + Phase 2 LLM reasoning) on a controlled escalation scenario.

6.1 Experimental Setup

Hardware: Apple Mac M4 with 10-core CPU, 10-core GPU, and 16-core Neural Engine; 48GB unified memory.

Software: Python 3.10, PyTorch 2.1 with MPS acceleration, CrowdSense v1.0

Model Configuration:

- DM-Count with VGG19 backbone, pretrained on NWPU-Crowd dataset.
- Rule engine with configurable thresholds (e.g., sparse_max=50, moderate_max=200, crowded_max=500).
- LLM deployment mode (both supported):
    - cloud-based: OpenAI GPT-5.2-chat via LiteLLM API
    - local: Ollama qwen2.5:14b
- Temporal window: 6 frames for Phase 2 risk assessment.

Backbone Reference Performance (DM-Count):

The following results are reported from the original DM-Count paper [1] and represent the pretrained backbone's performance on standard crowd-counting benchmarks. These numbers are provided to contextualize the perception backbone and are not CrowdSense system-level measurements

Method	UCF-QNRF MAE	UCF-QNRF RMSE	SHA MAE	SHA RMSE	SHB MAE	SHB RMSE
MCNN	277.0	426.0	110.2	173.2	26.4	41.3
CSRNet	120.3	208.5	68.2	115.0	10.6	16.0
DM-Count	**85.6**	**148.3**	**59.7**	**95.7**	**7.4**	**11.8**

As reported in [1], DM-Count achieves 29% lower MAE than CSRNet on UCF-QNRF, demonstrating strong performance on highly congested scenes

	Backbone	Validation set		Test set		
		MAE	RMSE	MAE	RMSE	NAE
MCNN [60]	FS	218.5	700.6	232.5	714.6	1.063
CSR net [20]	VGG-16	104.8	433.4	121.3	387.8	0.604
PCC-Net-VGG [10]	VGG-16	100.7	573.1	112.3	457.0	0.251
CAN [27]	VGG-16	93.5	489.9	106.3	**386.5**	0.295
SCAR [11]	VGG-16	81.5	397.9	110.0	495.3	0.288
Bayesian Loss [31]	VGG-19	93.6	470.3	105.4	454.2	0.203
SFCN [50]	ResNet-101	95.4	608.3	105.7	424.1	0.254
DM-Count (proposed)	VGG-19	**70.5**	**357.6**	**88.4**	388.6	**0.169**

Table 2: **Results of various methods on the NWPU validation and test sets.**

6.2 Datasets and Evaluation Inputs

CrowdSense is designed for periodic sampling in operational monitoring settings. In this project, the end-to-end system was evaluated using a controlled synthetic escalation sequence intended as an initial behavioral validation of the two-phase logic.

6.2.1 Initial Behavioral Validation (Synthetic Sequence)

The system was exercised using a controlled 12-frame sequence constructed by progressively tiling a high-resolution crowd-gathering image. This sequence was used to validate Phase 1 threshold behavior and the gated Phase 2 trigger logic across a complete lifecycle: Sparse buildup (frames 1–6), rapid surge to peak density (frames 7–10), and a final high-density compression-like pattern (frames 11–12). Each frame was processed through the full pipeline (DM-Count inference ⟶ feature extraction ⟶ Phase 1 classification), with feature records persisted to datasets/stats/

6.2.2 Planned Temporal Validation (SIMCD Integration)

To quantify the reliability of Phase 2 temporal reasoning beyond the controlled synthetic sequence, future evaluation will utilize the SIMCD framework [5], which provides simulated temporal crowd sequences with labeled abnormal and high-severity transitions. This will enable systematic measurement of (i) detection lead time - how early the system produces HIGH/CRITICAL risk assessments before labeled dangerous transitions - and (ii) trigger behavior quality, including how often Phase 2 produces Locking Alerts in scenarios that exhibit congestion-like patterns versus dense but stable flow

Alert System Evaluation:

As an initial behavioral validation, each frame was processed through the full CrowdSense pipeline (DM-Count inference ⟶ feature extraction ⟶ Phase 1 classification), with per-frame feature records persisted to datasets/stats/. The sequence spans an estimated count range of 4 to 581 persons, with entropy ranging from 2.30 to 6.27 and anisotropy from 0.74 to 0.96. The sequence is structured to cover all major system behaviors within a single continuous run:

- Sparse phase (frames 1–6, count 4–25): Low density, Phase 1 classifies as Sparse, Phase 2 not triggered
- Buildup and surge (frames 7–8, count 67–111): Phase 1 escalates to Moderate, Phase 2 triggers with sliding-window temporal analysis
- Peak density (frames 9–10, count 234–581): Phase 1 reaches Crowded / Heavily Crowded, Phase 2 detects surge escalation to CRITICAL
- Partial dispersal with compression (frames 11–12, count 335–286): Count decreases but entropy drops (6.24→5.71) while anisotropy rises (0.84→0.88), triggering a Locking Alert

Synthetic Sequence Feature Summary (12 Frames)

Frame	Count	Entropy	Aniso.	P1 Abs	n1 Abs	Hotspot	Phase 1 Status
tile_01	4	2.30	0.90	4.34	2.74	R4C3	Sparse
tile_02	7	2.91	0.92	7.28	2.63	R4C3	Sparse
tile_03	8	2.93	0.96	8.42	3.35	R4C3	Sparse
tile_04	16	3.44	0.89	14.02	3.82	R4C3	Sparse
tile_05	22	3.82	0.92	15.73	3.47	R4C3	Sparse
tile_06	25	4.11	0.87	15.30	4.15	R4C3	Sparse
tile_07	67	5.18	0.83	16.62	11.49	R5C8	Moderate
tile_08	111	5.62	0.82	18.77	14.09	R5C8	Moderate
tile_09	234	6.27	0.74	22.13	19.99	R5C4	Crowded
tile_10	581	6.10	0.86	61.54	65.64	R4C1	Heavily Crowded
tile_11	335	6.24	0.84	34.26	33.63	R5C8	Crowded
tile_12	286	5.71	0.88	37.78	37.41	R5C4	Crowded

Phase 2 Invocation:

LLM analysis (Phase 2) was triggered on 6 of 12 frames (frames 7–12). Each call received a sliding window of the 6 most recent feature records. Both GPT-5.2 (cloud, via LiteLLM) and Ollama qwen2.5:14b (local) were evaluated on identical input windows.

6.3 LLM's Comparisons

Alert System Comparison Axes:

- **Phase 1 only (Rule Engine)**: Deterministic threshold-based classification without temporal LLM analysis
- **Phase 1 + Phase 2 (GPT-5.2)**: Full two-phase system using OpenAI GPT-5.2-chat (cloud endpoint via LiteLLM)

- **Phase 1 + Phase 2 (Ollama qwen2.5:14b)**: Full two-phase system using locally hosted qwen2.5:14b (via Ollama)

Both LLM configurations received identical prompt instructions (lock-priority hierarchy) and identical input feature windows, enabling direct comparison of trigger differentiation quality, status assignment consistency, and response latency.

6.4 Metrics

Performance Metrics:

- **Phase 1 Latency**: End-to-end processing time per frame excluding LLM (ms)
- **Phase 2 Latency**: LLM response time per temporal assessment (ms)
- **Request Size**: Serialized input payload size (bytes)

Alert Quality Metrics:

- **Status Classification**: Phase 1 status label correctness against rule engine thresholds
- **Trigger Differentiation**: Whether the LLM correctly identified the primary escalation mechanism (Locking / Surge / Volume) per the prompt's lock-priority hierarchy
- **Escalation Tracking**: Whether the LLM correctly tracked status transitions (escalation / stable / de-escalation) across successive windows
- **Actionability**: Qualitative assessment of whether recommended actions are specific, grounded in observed data, and operationally useful

6.5 Results

6.5.1 CrowdSense Pipeline Evaluation

CrowdSense integrates DM-Count as its density estimation backbone. Our evaluation focuses on end-to-end pipeline behavior - Phase 1 status classification, Phase 2 trigger differentiation, and computational performance - rather than re-benchmarking the crowd counting model. We evaluate the system using the 12-frame synthetic escalation sequence described in 6.2, with Phase 2 assessed using both GPT-5.2 (cloud) and (vs) Ollama qwen2.5:14b (local).

6.5.2 Computational Performance

End-to-end latency measured on Mac M4 across the 12-frame evaluation sequence

Stage	Latency (ms)	Notes
Image preprocessing	15	Resize, normalize, tensor conversion
DM-Count inference	180	MPS-accelerated (vs ~3000ms CPU)
Feature extraction	8	NumPy statistical operations
Rule engine	<1	Threshold comparisons
Total (Phase 1)	**~205**	**~4.9 FPS capability**

GPT-5.2 API call	11,551–16,862	Mean: 15,177ms (N=6, remote)
Ollama qwen2.5:14b	52,388–56,894	Mean: 54,812ms (N=6, local)

Phase 2 LLM Performance Detail

Call #	Window	GPT-5.2 (ms)	Ollama (ms)	Request Size
1	tiles 2–7	13,995	52,388	~13.2 KB
2	tiles 3–8	16,088	56,809	~13.2 KB
3	tiles 4–9	11,551	55,544	~13.3 KB
4	tiles 5–10	16,569	54,502	~13.3 KB
5	tiles 6–11	15,998	52,735	~13.3 KB
6	tiles 7–12	16,862	56,894	~13.4 KB

Phase 1 achieves near real-time performance (~4.9 FPS), providing sufficient headroom for periodic sampling (e.g., one frame every 15 seconds) across multiple feeds. Phase 2 adds significant latency: GPT-5.2 averages ~15.2 seconds per assessment (cloud, network-dependent), while local Ollama inference averages ~54.8 seconds - approximately 3.6× slower. Request sizes remain consistent (~13.2–13.4 KB) across windows, confirming that the 6-frame sliding window produces a stable payload regardless of crowd density. The gated invocation strategy (Phase 2 only when status ≥ Moderate) ensured that only 6 of 12 frames incurred LLM latency.

6.5.3 Alert System Behavioral Validation

Phase 1 Classification Results: The rule engine correctly classified all 12 frames according to the configured thresholds (sparse_max=50, moderate_max=200, crowded_max=500). The transition from Sparse (frames 1–6) to Moderate (frames 7–8) to Crowded (frames 9, 11, 12) to Heavily Crowded (frame 10) directly reflects the count trajectory and threshold boundaries. No misclassifications occurred, as Phase 1 is fully deterministic.

GPT-5.2 Phase 2 Assessments
GPT-5.2 correctly differentiated three trigger types following the lock-priority hierarchy:

- **Surge Alert** (windows 1-4): Identified rapid count growth (>20% delta) as the primary escalation mechanism, correctly prioritizing surge over baseline volume
- **Volume Alert** (window 5): Recognized that count was declining but absolute volume remained at CRITICAL level, appropriately switching from surge to volume-based assessment
- **Locking Alert** (window 6): Detected entropy compression (6.10⟶5.71, drop of 0.39 >0.3 threshold) with sustained high anisotropy (≥0.84) across the last 3 frames, correctly invoking the highest-priority override per the prompt hierarchy

All 6 assessments referenced specific grid cells, provided actionable operator recommendations (e.g., deploy stewards, halt upstream inflow, open lateral egress), and included compact summaries of recent metrics consistent with the structured output schema in the prompt configuration.

6.6 Ablation / Sensitivity

6.6.1 Threshold Sensitivity

The rule engine thresholds control Phase 1 classification and, through the gating mechanism, determine when Phase 2 is invoked. Using the 12-frame sequence, we analyze how alternative threshold configurations affect system behavior. With the default configuration, the system triggers Phase 2 at frame 7 (count=67), providing early warning during the initial buildup. A more conservative setting delays Phase 2 until frame 9 (count=234), missing two surge windows. A more aggressive setting triggers Phase 2 one frame earlier, at the cost of an additional LLM call on a low-density frame (count=25) that yields limited analytical value.

6.6.2 Feature Contribution to Trigger Differentiation

The multi-feature representation enables the LLM to distinguish between escalation mechanisms that raw count alone cannot differentiate. Analysis of the 6 GPT-5.2 assessments reveals the role of each feature family:

- **Entropy** was decisive for Locking Alert detection in window 6. Without entropy, the system would see only a count decrease (581→335→286) and assign a de-escalating Volume status. The entropy drop (6.10→5.71) signaled spatial compression, enabling the highest-priority override.
- **Anisotropy** served as a confirming indicator: sustained values ≥0.82 across all 6 windows indicated directional flow constraint, but anisotropy alone did not trigger any status change. Its value was in combination with entropy for locking detection (threshold: anisotropy >0.80).
- **Count delta** was the primary driver for 4 of 6 assessments (Surge Alerts), with deltas ranging from +42 to +347 persons between successive frames.
- **n1_absolute** (densest grid cell) provided localization specificity: GPT-5.2 cited peak cell densities (e.g., n1=61.54 at tile_10, n1=37.78 at tile_12) in its actionable insights, enabling cell-specific operator recommendations.

6.7 Locking Detection Micro-Study

The most analytically significant Phase 2 assessment is the Locking Alert in window 6 (tiles 7–12), illustrating how temporal feature trends can reveal compression-like dynamics that a volume-only analysis would miss. After peaking at 581 persons (tile_10), the crowd count decreased across frames 11–12 (335→286). A volume-only system would interpret this as de-escalation; however, the feature trajectory indicates a potentially risky spatial reconfiguration

Frame	Count	Entropy	Anisotropy	n1 Abs	Interpretation
tile_10	581	6.10	0.86	65.64	Peak volume, moderate spread
tile_11	335	6.24	0.84	33.63	Count drops, entropy briefly rises
tile_12	286	5.71	0.88	37.41	Count drops further, entropy compresses, anisotropy rises

Locking condition met: entropy decreased by 0.39 (6.10→5.71) across the last 3 frames while anisotropy remained above 0.80 (0.86→0.84→0.88), satisfying the lock detection override (entropy drop >0.3 AND anisotropy >0.80).

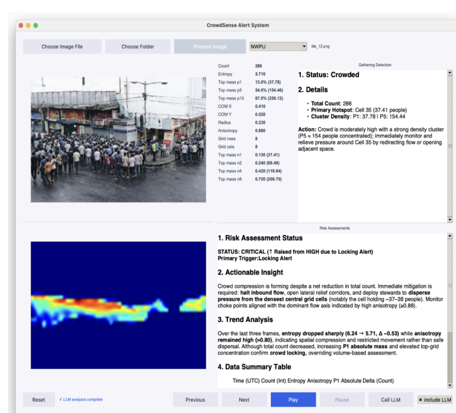
GPT-5.2 identified this pattern and escalated the risk level to CRITICAL with Locking Alert as the primary trigger, recommending pressure-relief measures such as opening lateral egress and halting upstream inflow.
This case demonstrates that density-derived features (entropy + anisotropy) can serve as proxies for compression-like dynamics even without explicit motion estimation. As discussed in 4.3.3, these indicators suggest constrained flow but do not confirm physical crush conditions; operators should verify via camera review or on-ground personnel before escalating response.

6.8 Cloud vs. Local LLM Comparison (GPT-5.2 vs. Ollama qwen2.5:14b)

To evaluate the trade-off between cloud and local LLM deployment, both GPT-5.2 (remote) and qwen2.5:14b (local on Mac M4) processed identical input windows using the same prompt (v4.4 lock-priority hierarchy).

Trigger Differentiation: GPT-5.2 correctly identified three trigger types (4× Surge, 1× Volume, 1× Locking) following the lock-priority hierarchy. Ollama classified all 6 windows as Volume Alert, failing to detect surge dynamics or the locking pattern in window 6. This indicates a qualitative gap: the local model applied the baseline volume rule but did not reliably apply the higher-priority override conditions.

Status Level Consistency: GPT-5.2 produced a monotonically escalating trajectory (MODERATE → MODERATE → HIGH → CRITICAL → CRITICAL → CRITICAL) consistent with the sequence's escalation profile. Ollama's trajectory (HIGH → HIGH → MODERATE → CRITICAL → CRITICAL → HIGH) contained inconsistencies, including a MODERATE classification for window 3 (count=234, n1_abs=22.13) despite exceeding the HIGH volume threshold, and a de-escalation to HIGH in window 6 despite meeting the locking override condition.



Actionable Quality: GPT-5.2 assessments referenced specific grid cells, cited feature thresholds and trends, and recommended targeted interventions (e.g., open lateral egress on a specified flank, deploy stewards to break anisotropic lanes). Ollama assessments were more generic, offering less localization and weaker feature-grounded justification

Latency Trade-off:

Metric	GPT-5.2 (Cloud)	Ollama qwen2.5:14b (Local)
Mean response time	15,177 ms	54,812 ms
Min / Max	11,551 / 16,862 ms	52,388 / 56,894 ms
Latency ratio	1.0×	3.6×
Endpoint	Remote (API)	Local (Mac M4)
Request size	~13.3 KB	~13.3 KB

Summary: GPT-5.2 delivers substantially better trigger differentiation, more consistent status tracking, and more actionable recommendation s at approximately one-third the latency of local Ollama inference. The local model may be suitable for privacy-sensitive or offline deployments where Volume-level alerting is sufficient, but it does not currently match the cloud model's ability to apply the full lock-priority reasoning hierarchy. Future work could explore fine-tuning the local model on the specific prompt structure to improve trigger differentiation.

---

7. Discussion and Future Work

7.1 Model Enhancements

- **Backbone upgrade**: Replace VGG19 with modern architectures (EfficientNet, ConvNeXt)
- **Attention mechanisms**: Add self-attention for long-range spatial dependencies
- **Multi-scale fusion**: Improve accuracy across varying crowd densities
- **Uncertainty quantification**: Confidence intervals for count estimates

7.2 System Extensions

- **Video processing**: Temporal analysis for crowd flow estimation
- **Real-time streaming**: Live camera feed integration
- **Multi-camera fusion**: Aggregate views from multiple angles
- **Edge deployment**: Optimized models for embedded devices

7.3 Analysis Capabilities

- **Crowd flow vectors**: Direction and velocity estimation
- **Anomaly detection**: Unusual pattern identification

- **Predictive alerts**: Trend-based early warning
- **Comparative analytics**: Cross-scene and historical analysis

7.4 Integration Options

- **Web interface**: Browser-based deployment
- **REST API**: Headless processing service
- **Mobile application**: On-device inference
- **Dashboard integration**: Enterprise monitoring systems

7.5 Limitations

The current CrowdSense implementation has several limitations that should be considered when interpreting results or planning deployments:

- Domain shift sensitivity: The DM-Count backbone is pretrained on standard crowd counting datasets. Performance may degrade on scenes with significantly different camera angles, lighting conditions, or crowd demographics without domain-specific fine-tuning.

- Viewpoint and threshold dependence: Detection thresholds are calibrated for typical surveillance perspectives. Extreme camera angles, fisheye distortion, or unusual scene geometry may require threshold recalibration to avoid false alarms or missed detections.

- Evaluation scope: System behavior was validated using a controlled synthetic escalation sequence. Further evaluation on real CCTV environments (and, where feasible, incident-labeled data) is required to quantify generalization and operational reliability.

- Phase 2 latency constraints: Phase 2 runs as an asynchronous assessment service and introduces approximately 11–56 seconds of latency (cloud vs. local). In fast-evolving incidents, this delay may exceed the window for effective intervention; CrowdSense should therefore be positioned as an early-warning tool for detecting buildup and emerging compression patterns rather than a real-time tactical control loop.

- LLM cost and scalability: Cloud-based Phase 2 introduces recurring API costs that scale with the number of triggered assessments and monitored feeds. In our configuration, the cloud cost is approximately $1 per 100 Phase 2 requests, so frequent triggers in high-traffic deployments can increase operational cost, motivating gated invocation, caching, and scenario-specific threshold tuning.

7.6 Ethics and Privacy Considerations

CrowdSense is designed with privacy-conscious principles appropriate for public safety monitoring:

- **No identity recognition**: The system operates on density maps and aggregate statistics, not individual identification. No face recognition, re-identification, or personal tracking is performed.

- **Feature storage over video**: Where possible, deployments should store extracted feature records rather than raw video, minimizing privacy exposure while preserving analytical capability.
- **Access control and logging**: Production deployments should implement role-based access controls and audit logging to ensure accountability for system use and alert responses.
- **Bias considerations**: Camera placement, lighting conditions, and scene composition can affect density estimation accuracy across different crowd demographics. Operators should be aware that systematic biases in camera coverage may lead to uneven detection sensitivity across monitored areas.

---

8. References

[1] Wang, B., Gao, J., Xu, W., & Shen, C. (2020). **Distribution matching for crowd counting**. *Advances in Neural Information Processing Systems (NeurIPS)*. (NeurIPS Proceedings)

[2] Li, Y., Zhang, X., & Chen, D. (2018). **CSRNet: Dilated convolutional neural networks for understanding the highly congested scenes**. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. (CVF Open Access)

[3] Helbing, D., Johansson, A., & Zein Al-Abideen, H. (2007). **The dynamics of crowd disasters: An empirical study**. *Physical Review E, 75*(4), 046109. (arXiv)

[4] Bek, S., & Monari, E. (2016). **The crowd congestion level: A new measure for risk assessment in video-based crowd monitoring**. In *2016 IEEE Global Conference on Signal and Information Processing (GlobalSIP)*. (ResearchGate)

[5] Sedky, M., Bamaqa, A., Bosakowski, T., Bastaki, B. B., & Alshammari, N. (2022). **SIMCD: SIMulated crowd data for anomaly detection and prediction**. *Expert Systems with Applications, 203*, 117475. (ScienceDirect)

[6] Simonyan, K., & Zisserman, A. (2015). **Very Deep Convolutional Networks for Large-Scale Image Recognition**. *International Conference on Learning Representations (ICLR)*. (arXiv)
